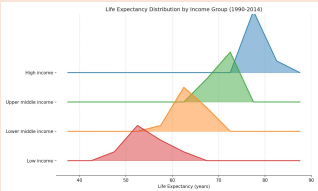
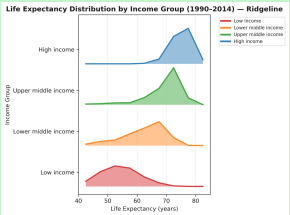
DV-Evol Bad Case python：Visual mapping failure

Question: Generate a ridgeline plot comparing life expectancy (1990–2014) across income groups. Use fixed 5-year bins from 40 to 85, normalize to percent frequency, and stack the ridges from "Low income" to "High income" using designated HEX colors.

Gemini-3-pro-preview:



Ground Truth:



Plot code:

```
df_plot = pd.read_csv('result.csv')
colors = {"Low income": "#d62728", "Lower middle income":
"#ff7f0e",
        "Upper middle income": "#2ca02c", "High income":
"#1f77b4"}
bin_centers = np.linspace(42.5, 82.5, 9)
x_coords = np.concatenate(([37.5], bin_centers, [87.5]))

fig, ax = plt.subplots(figsize=(10, 6))
for i, group in enumerate(groups):
    base_y = i * 1.0  # Vertical offset
    pcts = df_plot[df_plot['Income Group'] ==
group]['Percentage'].values
    y_coords = base_y + np.concatenate(([0], pcts * 0.025, [0]))
# Scaled heights

    # Render ridge
    ax.fill_between(x_coords, base_y, y_coords,
color=colors[group], alpha=0.5, zorder=10-i)
    ax.plot(x_coords, y_coords, color=colors[group], lw=2,
zorder=10-i)

# Formatting
ax.set_yticks([i * 1.0 for i in range(len(groups))])
ax.set_yticklabels(groups)
ax.set_xlabel("Life Expectancy (years)")
ax.set_title("Life Expectancy Distribution (1990-2014)")
ax.spines[['top', 'right', 'left']].set_visible(False)
plt.savefig('result.png', dpi=150)
```

Error Analysis: 1. Data Integrity

✅1.1 Trend/Pattern: 1/1 The distribution curves successfully mirror the multi-modal trends of the Ground Truth.

❌1.2 Completeness: 0/1 Critical data series or specific years are missing from the visualization.

✅1.3 Label Accuracy: 1/1 Axis tick values and labels do not precisely match the original data points.

❌1.4 Data Consistency: 0/1 Proportional relationships between income groups remain accurate across the X-axis.

2. Style Imitation

✅2.1 Color Palette: 1/1 The model correctly applied the designated HEX codes for all income categories.

❌2.2 Chart Elements: 0/1 Specific ridgeline outlines and gridline styles were not properly reproduced.

✅2.3 Background: 1/1 The background color and light/dark mode are consistent with the Ground Truth.

❌2.4 Font and Text Styles: 0/1 Font weight and sizing for titles and labels deviate from the original style.

✅2.5 Element Spacing: 1/1 Distances between labels and the plot area are uneven, creating a cluttered look.

❌2.6 Borders and Shadows: 0/1 Visual effects like drop shadows or specific border strokes were omitted.

3. Layout & Aesthetics

❌3.1 Labels: 0/1 Legend and title positioning are misaligned compared to the standard layout.

✅3.2 Readability: 1/1 Excessive ridge overlap and small font sizes compromise the overall legibility.

❌3.3 Axis Alignment and Spacing: 0/1 The axes are poorly calibrated, causing data to shift away from intended baselines.

❌3.4 Element Alignment: 0/1 Graphical components fail to align precisely with their respective axis ticks.

❌3.5 Proportionality and Aspect Ratio: 0/1 An incorrect aspect ratio has led to vertical distortion of the ridge shapes.

❌3.6 Overall Balance: 0/1 The composition lacks visual equilibrium, making the chart difficult to interpret quickly.

Score: 6/16